# Lab 04

## A sustainable communities evidence brief

Teaching Week 04 | Individual work. Submit in R or Python.

**Purpose**

You will turn an open dataset into a short evidence brief for a public audience. The brief must be reproducible, trace each claim to a table or figure, and distinguish what the data show from what they cannot establish.

## Learning outcomes

After completing this lab, you should be able to

- formulate a focused public-interest data question;
- organize a reproducible report around evidence rather than workflow;
- integrate tables, figures, citations, and limitations; and
- communicate technical choices in plain language.

## Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

**Option A | Low Carbon Communities Grant Recipients**

**Publisher.** Nova Scotia Department of Energy
**Official record.** https://data.novascotia.ca/d/p9y5-d99g
**Record identifier.** `p9y5-d99g`
**Licence and format.** Nova Scotia Open Government Licence; CSV
**Expected variables.** fiscal year; funding round; organization; category; project name; amount awarded; area
**Use in this lab.** build an evidence brief about low-carbon investment
**Why it fits.** Project records support a reproducible question with local public relevance.

Export CSV from the catalogue. One row is a funded project, not a resident or a measured emissions reduction. Use `Amount Awarded by Program` in Canadian dollars. Every published `Fiscal year` label is a range, and 2018–2020 overlaps 2019–2020; keep the labels as published when comparing funding rounds. Check category and area spelling before grouping. A province-wide project is not a community-specific project, and an award does not demonstrate its eventual environmental effect.

**Option B | Tourism Nova Scotia Visitor Exit Survey Communities**

**Publisher.** Nova Scotia Communities, Culture, Tourism and Heritage
**Official record.** https://data.novascotia.ca/d/is4a-t3qd
**Record identifier.** `is4a-t3qd`
**Licence and format.** Nova Scotia Open Government Licence; CSV
**Expected variables.** year; community; region; proportion of surveyed visitors
**Use in this lab.** build an evidence brief about community activity
**Why it fits.** Community indicators support concise reporting with explicit limitations.

Export the CSV with `Year`, `Community`, `Region`, and `Percent`. Each row is a community-year survey estimate. The field named `Percent` stores proportions. Verify and document the units used in your reported quantities. The survey covers non-resident overnight visitors, including communities they visited briefly. People can visit several communities, so these percentages do not form parts of a 100% total. Compare communities or years without summing those overlapping estimates. Keep an absent community-year distinct from zero, and document changes in region labels. This extract has no sample-size or standard-error column.

**Option C | Commercial Fisheries - Licences - Atlantic Region Licences**

**Publisher.** Fisheries and Oceans Canada
**Official record.** https://open.canada.ca/data/en/dataset/2cc50279-633c-4b60-8037-a642fe5ab963
**Record identifier.** `2cc50279-633c-4b60-8037-a642fe5ab963`
**Licence and format.** Open Government Licence – Canada; CSV or XLS
**Expected variables.** year; province; DFO region; species or registration category; count
**Use in this lab.** build an evidence brief about coastal livelihoods
**Why it fits.** Licence distributions connect reproducible analysis to Atlantic communities.

Use one table family and state its unit. Species licences, registered harvesters, and registered vessels are different quantities. The current DFO Atlantic Region Licences page provides annual HTML tables and XLS downloads if a catalogue attachment is unavailable. Prefer a catalogue CSV. If you use a workbook, keep the original file, export the one sheet you need to a CSV with a single header row in a spreadsheet program, and read that CSV as in earlier labs; reading XLS files in code is not required. Record the sheet, the rows kept, how stacked header rows became column names, and any table notes that qualify the counts. Compare the same category across regions or years. Keep totals separate from their components, and do not interpret licences as fish abundance, landings, employment, or distinct people. Community-level detail is not promised by these tables.

## Data contract

Choose one listed dataset and prepare a short reproducible evidence brief.

- State the observational unit and retain enough observations to support at least two defensible comparisons. Report the number of rows behind each comparison; do not duplicate rows to enlarge the dataset.
- Include one numeric measure and one community, region, project, sector, species, or time variable.
- Reuse summaries, cleaning decisions, and statistical graphics from earlier weeks.
- State one focused public-interest question and one bounded conclusion.

- Cite the official record and document every data-preparation decision. A plain Markdown reference section, as shown in the Class 08 notes, is sufficient.
- Include one limitation that materially constrains the conclusion.
- Render the report from source in either R Markdown or a Python notebook, using the Class B workflow.

**What the contract does not settle**

These requirements make the evidence traceable. They do not establish that the records represent a wider population or that a program caused an observed difference.

## Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I stated one public-interest question and identified the intended audience.
- ☐ My table satisfies every rule in the data contract.
- ☐ I can reuse prior summaries, cleaning, and visualization methods.
- ☐ My central claim is bounded by the available evidence.
- ☐ I identified one material limitation.
- ☐ I can render the brief reproducibly in either R or Python.

## Cumulative method

This lab carries forward the full Week 01 to Week 04 path from observational-unit definition through cleaning, table manipulation, statistical visualization, and reproducible reporting. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

## Work program

### 1. Frame the brief

Name the intended audience, the decision or question, and the scope. Treat your proposed answer as provisional until you have examined the evidence.

### 2. Build the evidence chain

Produce one compact table and two complementary figures from a scripted cleaning and analysis pipeline. Reuse the Class A grouping, transformation, or join methods when needed. Check join keys and row counts before summarizing. Keep every reported quantity tied to an object computed by the report source.

### 3. Write for the audience

Lead with the answer supported by your results, explain quantities in plain language, and place methodological detail where it helps a reader evaluate the claim. Give figures units, captions, and accessible labels. Compute numerical statements with inline R code or Python-generated text rather than typing a second copy of the result.

### 4. Bound the conclusion

Separate description from causal explanation. State one material limitation, one plausible alternative explanation, and one useful next data source. Do not infer program effectiveness, population totals, or statistical significance from data that lack the required outcomes, denominators, or uncertainty information.

### 5. Prove reproducibility

Render the final PDF from a clean session and provide the exact command and data-access instructions. Check the rendered table, figures, numerical statements, citations, and page breaks. A successful render alone does not establish that the analysis is correct.

## Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Downloaded data and generated results are not course source, so they never go into the course repository. Commit the report, its source, and the generated tables and figures it cites to your private work repository. Keep that evidence private until the deadline, and publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

## Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-04/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on October 8, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-04/`. A branch URL or movable tag alone is not a fixed submission. The package includes

- a concise PDF report that answers the analytical question, together with the completed selection checklist;

- the complete R or Python source plus the generated tables, figures, and any summary statistics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

## Assessment standard

The course uses one rubric across all integrated labs. The best eight of the nine weekly lab results contribute 15% of the course grade.

**Table 1.** Integrated lab marking criteria.

| Criterion | Weight (%) |
|---|---|
| Dataset provenance and suitability | 15 |
| Reuse of earlier methods | 20 |
| Current-week analytical method | 35 |
| Interpretation and limitations | 20 |
| Reproducibility and communication | 10 |
| Total | 100 |

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.